```
          "type":"keep",
          "keep_words":["alpha", "beta", "gamma"]
        }
      }
    }
  }
}
```

Keep words; all other words are discarded.

This code is a bit of a handful, but it's easy to understand. In the first part, where we define a custom analyzer, we provide a list of filters (both character and token filters, if needed) and a tokenizer. You can imagine this as the entry point to the analyzer definition.

The second part of the code defines the filters declared earlier. For example, `greek_symbol_mapper`, redeclared in the new `char_filter` section, uses `mapping` as the filter's type with a set of mappings. The same goes for the `filter` block, which defines the `keep_words` filter. This filter removes any words that aren't in the `keep_ words` list.

Now we can execute a test sample for analysis. The sentence in the following listing should be passed through the test analysis phase.

Listing 7.26 Parsing Greek letters from normal text

```
POST index_with_parse_greek_letters_custom_analyzer/_analyze
{
  "text": "α and β are roots of a quadratic equation. γ isn't",
  "analyzer": "greek_letter_custom_analyzer"
}
```

Greek letters (α, β, and γ, for example) are processed by the custom analyzer (`greek_letter_custom_analyzer`), which output the following:

```
"alpha","beta","gamma"
```

The rest of the words, like *roots* and *quadratic equation*, are removed.

So far, we have looked at the analyzers in detail, including built-in and custom ones. We can configure analyzers not just at the field level but also at the index or query level. And we can also specify a different analyzer for search queries if our requirements dictate. We discuss these points in the next section.

7.5 Specifying analyzers

Analyzers can be specified at the index, field, and query level. Declaring analyzers at the index level provides an index-wide default catch-all analyzer for all text fields. However, if further customization is required on a field level, we can also enable a different analyzer at that level. In addition, we can provide a different analyzer instead of the index-time analyzer while searching. Let's look at each of these options.

7.5.1 Analyzers for indexing

Sometimes we may have a requirement to set different fields with different analyzers—for example, associating a name field with a `simple` analyzer and a credit card